

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

___show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ___S0/0/0___

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? ___129___

Does R2 show up as an OSPF neighbor on R1? ___Yes___

Does R2 show up as an OSPF neighbor on R3? ___No___

What does this information tell you?

___All traffic from R3 to the 192.168.2.0/24 network will be forwarded via R1. Because the S0/0/1 interface on R2 is still configured as a passive interface, OSPF cannot advertise routing information through this interface, preventing R3 from establishing an OSPF neighbor relationship with R2. On this path, the accumulated cost for R3 to reach the destination network is 129. This is calculated as follows: the traffic must traverse two T1 serial links from R3 (each with a cost of 64), plus the Gigabit 0/0 LAN link on R2 (with a cost of 1), totaling 129.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

___R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65. One T1 serial link cost 64 and R2 Giabit 0/0 LAN link cost 1.

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The default 100 Mb/s reference bandwidth is outdated. It makes all links faster than 100 Mb/s (like Gigabit Ethernet) have the same cost. Increasing it allows for accurate cost calculations on modern high-speed links.

Step 2: Change the bandwidth for an interface.

- g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

- Cost to 192.168.3.0/24: R1 S0/0/1 (781) + R3 G0/0 (1) = 782
- Cost to 192.168.23.0/30: R1 S0/0/1 (781) + R3 S0/0/1 (64) = 845

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

Each serial link now has a cost of 781. Since the route to 192.168.23.0/24 traverses two serial links, the total cost is $781 + 781 = 1,562$.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF selects the path with the lowest cumulative cost. The route via R1-S0/0/0, R2-S0/0/1, and R3-G0/0 has a total cost of $781 + 781 + 1 = 1,563$. This is lower than the alternative route through R1-S0/0/1 and R3-G0/0, which totals $1,565 + 1 = 1,566$.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

It is important to control the OSPF Router ID to ensure network stability and predictable DR/BDR elections. If the Router ID is tied to a physical interface, it can change when that interface goes down, causing disruption. To prevent this, the Router ID should be manually configured using the router-id command or assigned from a persistent loopback interface.

2. Why is the DR/BDR election process not a concern in this lab?

DR/BDR elections are only performed on multi-access network types (e.g., Ethernet) to manage link-state advertisements efficiently. This lab uses point-to-point serial links exclusively. On a point-to-point link, the concept of a DR/BDR does not apply, so the election process is irrelevant.

3. Why would you want to set an OSPF interface to passive?

Setting an interface to passive stops OSPF from sending Hello packets on that link, which prevents neighbor adjacencies from forming. This is useful for conserving bandwidth on LAN segments where you don't need OSPF communication, while still allowing the router to advertise that network to its other OSPF

neighbors.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

_____ OSPF supports **two-layer hierarchy** through the use of areas. By dividing a large network into multiple areas, OSPF limits the propagation of routing updates and reduces the size of the link-state database (LSDB). All areas must connect to **Area 0 (backbone area)**, which summarizes routes between areas, improving scalability and reducing processing and memory overhead on routers.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

_____ network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

_____ EIGRP has the lowest AD (90), it will be selected regardless of the metric values.
